

Technical Datasheet

DESCRIPTION

Terluran GP-22 is an easy-flow, general purpose injection molding grade with high resistance to impact and heat distortion; intended for a wide range of applications, particularly in the housings sector.

FEATURES

- Excellent colorability
- Medium flow
- Good impact resistance
- Good heat distortion resistance
- High quality surface finish and gloss

APPLICATIONS

- Injection molding
- Appliance housings
- Household and sanitary appliances
- Toys
- Automotive components

Property, Test Condition	Standard	Unit	Values
Rheological Properties			
Melt Flow Rate, 200 °C/5 kg	ASTM D 1238	g/10 min	1.5
Melt Volume Rate 220 °C/10 kg	ASTM D 1238	cm³/10 min	
Melt Flow Rate, 220 °C/10 kg	ASTM D 1238	g/10 min	19
Melt Volume Rate 230 °C/3.8 kg	ASTM D 1238	cm³/10 min	4.8
Mechanical Properties			
Izod Notched Impact Strength, 23°C (73°F)	ASTM D 256	ft-lb/in	5.6
Izod Notched Impact Strength, -18°C (0°F)	ASTM D 256	ft-lb/in	1.9
Izod Notched Impact Strength, -30°C (-22°F)	ASTM D 256	ft-lb/in	1.1
Izod Notched Impact Strength, 23 °C	ISO 180/A	kJ/m²	
Izod Notched Impact Strength, -30 °C	ISO 180/A	kJ/m²	
Charpy Notched Impact Strength, 23° C	ISO 179	kJ/m²	
Charpy Notched Impact Strength, -30° C	ISO 179	kJ/m²	
Charpy Unnotched, 23° C	ISO 179	kJ/m²	
Charpy Unnotched, -30° C	ISO 179	kJ/m²	
Tensile Stress at Yield, 23° C		psi	6520
Tensile Strain at Yield, 23° C	ASTM D 638	%	
Tensile Modulus	ASTM D 638	psi x 10³	334

Terluran GP-22

Acrylonitrile Butadiene Styrene (ABS)



Driving Success. Together.

Property, Test Condition	Standard	Unit	Values
Nominal Strain at Break, 23 °C		%	
Elongation, Failure	ASTM D 638	%	2.6
Flexural Strength	ASTM D 790	psi	9430
Flexural Modulus	ASTM D 790	psi x 10 ³	334
Hardness, Rockwell	ASTM D 785	R scale	103
Hardness, Ball Indentation			
Thermal Properties			
Vicat Softening Temperature VST/B/50 (50°C/h, 50N)	ISO 306	°F	
Vicat Softening Temperature, VST/A/50 (50°C/h, 10N)	ISO 306	°F	204
Heat Deflection Temperature A; (annealed, 1.8 MPa)		°F	
Heat Deflection Temperature B; (annealed, 0.45 MPa)		°F	
DTUL @ 264 psi - Unannealed	ASTM D 648	°F	172
DTUL @ 66 psi - Unannealed	ASTM D 648	°F	195
DTUL @ 264 psi - Annealed	ASTM D 648	°F	210
DTUL @ 66 psi - Annealed	ASTM D 648	°F	219
Coefficient of Linear Thermal Expansion	ASTM D 696	10 ⁻⁴ /°F	
Thermal Conductivity			
Electrical Properties			
Dielectric Constant at 106 CPS (1000000 Hz, 0,0394 in)	ASTM D 150	-	2.8
Dissipation Factor (100 Hz)		-	
Dissipation Factor (1 MHz)		-	
Relative Permittivity (100 Hz)		-	
Relative Permittivity (1 MHz)		-	
Volume Resistivity	ASTM D 257	-	>1E13
Surface Resistivity	ASTM D 257	-	
Other Properties			
Density	ASTM D 792	-	1.04
Water Absorption, Saturated at 23°C	ASTM D 570	%	1
Moisture Absorption, Equilibrium 23°C/50% RH		%	
Yellowness Index		-	

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Property, Test Condition	Standard	Unit	Values
Processing			
Linear Mold Shrinkage	ASTM D 955	in/in	0.004 - 0.007
Melt Temperature Range		°F	425 - 500
Mold Temperature Range		°F	85 - 140
Injection Velocity		in/s	8
Drying Temperature		°F	175
Drying Time		h	2 to 4

Typical values for uncolored products

SUPPLY FORM

Terluran® is delivered as spherical pellets. The bulk density of the pellets is from 0.55 to 0.65 g/cm³. Standard Packaging unit: 25 kg PE-bag on palette, shrunk or wrapped with PE film or delivery in silo trucks. PE bags should not be stored outside. In dry areas with normal temperature control, Terluran pellets can be stored for relatively long periods of time without any change in mechanical properties. Under poor storage conditions, Terluran absorbs moisture, but this can be removed by drying.

PRODUCT SAFETY

No adverse effects on the health of processing personnel have been observed if the products are correctly processed and the production areas are suitably ventilated. For styrene, acrylonitrile and 1,3-butadiene the maximum allowable workplace concentrations must be observed according to the pertaining national regulations. In Germany, the following limit values are valid (Oct. 2002): styrene, MAK-value: 20 ml/m³ = 86 mg/m³; acrylonitrile, TRK-value: 3 ml/m³ = 7 mg/m³ and 1,3-butadiene, TRK-value: 5 ml/m³ = 11 mg/m³. According to EU directive 67/548/EWG, Annex I and TRGS 905 (Oct. 2002), acrylonitrile and 1,3-butadiene are classified as carcinogenic, category 2 ('substances which should be regarded as if they are carcinogenic to man') and 1 (substances known to be carcinogenic to man), respectively. Experience has shown that during appropriate processing of Terluran with suitable ventilation the values obtained are well below the limits mentioned above. TRGS 402 (Germany) can be used for determining and assessing the concentrations of hazardous substances in the air within working areas. Inhalation of gaseous degradation products, such as those which may arise on severe overheating of the material or during pumped evacuation, must be avoided. Further information can be found in our Terluran safety data sheets.

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